

Customer Segmentation Analysis: Comparing K-Means and DBSCAN

1. Exploratory Data Analysis

We did **exploratory data analysis** using standard Python libraries like **NumPy** and **Pandas** for data manipulation, and **Matplotlib** for visualization. To ensure proper imputation, we utilized the **KNNImputer** from Scikit-Learn to address missing values in the income column and applied **standard scaling** to prepare the features for clustering.

The initial raw data, downloaded from kaggle, consisted of 2,240 records across **29 variables**, including demographics like birth year and education alongside behavioral data such as purchase history and campaign responses. We refined this data by creating more intuitive features, such as calculating customer age and loyalty time. We also simplified categorical variables, grouping marital status into broader categories and mapping education levels to a numerical scale to facilitate algorithmic processing.

Data cleaning was a critical step in this phase. We identified and removed several redundant or constant columns that lacked variance. Furthermore, we pruned the dataset of unrealistic **outliers**, specifically targeting extreme age entries and highly unusual income levels. By the end of the EDA process, we had consolidated the information into **23 core features**, which were then standardized and saved as a cleaned dataset for the modeling phase.

2. K-Means Clustering

Our first clustering approach utilized the **K-Means algorithm**. To determine the most effective number of segments, we evaluated several configurations ranging from 2 to 19 clusters. We used a combination of the **Elbow Method**, **Silhouette Scores**, and the **Davies-Bouldin Index** to find the "sweet spot" where the clusters were both distinct and compact.

The analysis revealed that dividing the customers into **five clusters** yielded the most meaningful results. While the Silhouette Score tended to decrease as the number of clusters grew, the Davies-Bouldin Index confirmed that **five clusters** provided the best balance of separation. The final model successfully partitioned the **2,235 customers** into five valid segments without leaving any noise points behind.

3. DBSCAN Clustering

Next, we explored **DBSCAN**, a **density-based clustering** method. Initially, we applied the algorithm to the full 23-dimensional feature space. By analyzing the distance between neighboring data points, we estimated an optimal search radius. However, the results were disappointing; the algorithm struggled with the **high dimensionality** of the data, essentially grouping almost every customer into one massive cluster while labeling a few dozen as **outliers**.

In an attempt to improve the outcome, we reduced the complexity of the data using **Principal Component Analysis (PCA)**, retaining about 43% of the original variance within **three components**. Even with this reduction and various parameter adjustments, **DBSCAN** continued to show a strong tendency toward a **single-cluster solution**. This suggests that the customer data lacks the specific density variations required for **DBSCAN** to identify multiple distinct groups effectively.

4. Comparative Benchmark and Conclusion

To definitively choose the best approach, we conducted a head-to-head comparison between the five-cluster **K-Means** model and the most promising **DBSCAN** configuration. While **DBSCAN** appeared to have a higher Silhouette Score, this was a mathematical byproduct of having only two groups (one being a very small set of **outliers**). **K-Means** proved superior across more robust metrics, showing significantly better scores in the Davies-Bouldin and **Calinski-Harabasz indices**.

In conclusion, **K-Means** is the clear winner for this specific marketing dataset. While **DBSCAN** is powerful for finding oddly shaped clusters, it failed to discover a meaningful multi-cluster structure in this data. **K-Means**, on the other hand, successfully segmented the audience into **five actionable categories**, making it the more reliable tool for understanding this customer base.